

## CASLAV, Czech Republic

WMO: 116240

Lat: 49.9408N

Lon: 15.3864E

Elev: 240

StdP: 98.48

Time Zone: 1.00 (EUC)

Period: 97-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |       |       |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|-------|-------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |       | 99%   |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB  | DP    | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)   | (g)   | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -12.3      | -9.6 | -15.5                         | 1.0 | -10.9 | -12.6 | 1.3 | -8.5 | 12.4                  | 6.9  | 11.1 | 5.4  | 1.1                   | 310  | 0.522 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 10.7                   | 31.0            | 20.2 | 29.1 | 19.6 | 27.3 | 18.8 | 21.4                | 28.4 | 20.5 | 27.2 | 19.7 | 25.7 | 3.2                  | 140  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 19.1                            | 14.3 | 23.8 | 18.3 | 13.5 | 22.8 | 17.5 | 12.9 | 22.1 | 63.1          | 28.3 | 60.0 | 27.5 | 57.1 | 25.9 | 25.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| 9.8               | 8.4  | 7.4 | -15.7                      | 34.1 | 3.8                | 1.8 | -18.5                                              | 35.4 | -20.7      | 36.4 | -22.9      | 37.4 | -25.6      | 38.7 |    |    |
|                   |      |     | -16.1                      | 23.1 | 3.8                | 0.8 | -18.8                                              | 23.7 | -21.0      | 24.1 | -23.2      | 24.6 | -25.9      | 25.2 |    |    |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 9.9    | -0.5  | 1.2   | 4.9   | 9.9   | 14.4  | 17.7  | 19.6  | 19.4  | 14.9  | 9.9   | 5.1   | 1.2  |
|                                                                                  | DBStd       | 8.10   | 5.17  | 4.79  | 4.14  | 4.16  | 3.68  | 3.50  | 3.20  | 3.24  | 3.39  | 3.90  | 3.88  | 4.37 |
|                                                                                  | HDD10.0     | 1270   | 325   | 246   | 164   | 52    | 6     | 0     | 0     | 0     | 2     | 51    | 152   | 272  |
|                                                                                  | HDD18.3     | 3267   | 583   | 479   | 418   | 254   | 131   | 53    | 24    | 25    | 111   | 262   | 397   | 530  |
|                                                                                  | CDD10.0     | 1219   | 1     | 1     | 5     | 49    | 142   | 231   | 296   | 292   | 151   | 47    | 5     | 0    |
|                                                                                  | CDD18.3     | 173    | 0     | 0     | 0     | 1     | 9     | 34    | 62    | 59    | 9     | 1     | 0     | 0    |
|                                                                                  | CDH23.3     | 1736   | 0     | 0     | 0     | 14    | 106   | 338   | 608   | 581   | 88    | 1     | 0     | 0    |
|                                                                                  | CDH26.7     | 523    | 0     | 0     | 0     | 1     | 18    | 90    | 207   | 191   | 16    | 0     | 0     | 0    |
| Wind                                                                             | WSAvg       | 3.0    | 3.5   | 3.4   | 3.5   | 3.1   | 2.9   | 2.6   | 2.7   | 2.4   | 2.6   | 2.8   | 3.1   | 3.5  |
| Precipitation                                                                    | PrecAvg     | 609    | 35    | 29    | 42    | 33    | 69    | 73    | 90    | 74    | 51    | 38    | 39    | 38   |
|                                                                                  | PrecMax     | 792    | 64    | 58    | 117   | 74    | 124   | 156   | 226   | 166   | 109   | 86    | 83    | 71   |
|                                                                                  | PrecMin     | 456    | 10    | 4     | 9     | 2     | 9     | 21    | 13    | 31    | 10    | 4     | 1     | 11   |
|                                                                                  | PrecStd     | 88     | 16    | 17    | 24    | 16    | 28    | 27    | 43    | 36    | 31    | 19    | 20    | 15   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 12.3  | 13.6  | 18.8  | 25.2  | 28.8  | 32.0  | 33.8  | 33.9  | 28.8  | 22.5  | 16.3  | 11.7 |
|                                                                                  |             | MCWB   | 9.5   | 9.2   | 11.4  | 15.5  | 18.3  | 21.3  | 20.9  | 20.3  | 19.0  | 15.7  | 12.9  | 9.4  |
|                                                                                  | 2%          | DB     | 10.1  | 11.5  | 15.9  | 22.4  | 26.2  | 29.4  | 31.1  | 30.7  | 25.8  | 19.7  | 14.1  | 9.8  |
|                                                                                  |             | MCWB   | 7.6   | 8.2   | 9.9   | 13.9  | 17.4  | 20.4  | 20.3  | 19.9  | 17.9  | 14.7  | 11.5  | 7.9  |
|                                                                                  | 5%          | DB     | 8.0   | 9.7   | 13.6  | 20.0  | 24.0  | 27.1  | 28.9  | 28.7  | 23.5  | 17.7  | 12.3  | 8.3  |
|                                                                                  |             | MCWB   | 6.1   | 7.0   | 8.9   | 12.9  | 16.2  | 18.9  | 19.7  | 19.4  | 16.9  | 14.0  | 10.2  | 6.5  |
|                                                                                  | 10%         | DB     | 6.1   | 7.6   | 11.4  | 17.8  | 21.8  | 25.1  | 26.8  | 26.6  | 21.4  | 15.9  | 10.7  | 6.9  |
|                                                                                  |             | MCWB   | 4.5   | 5.5   | 7.8   | 11.6  | 15.3  | 18.1  | 18.8  | 18.7  | 15.8  | 13.1  | 9.0   | 5.5  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 9.9   | 9.8   | 12.2  | 16.1  | 20.0  | 22.8  | 22.4  | 22.1  | 19.8  | 16.7  | 13.5  | 9.7  |
|                                                                                  |             | MCDB   | 12.2  | 12.7  | 17.0  | 23.5  | 26.6  | 30.1  | 29.8  | 29.0  | 26.6  | 20.3  | 15.5  | 11.3 |
|                                                                                  | 2%          | WB     | 7.8   | 8.5   | 10.6  | 14.7  | 18.2  | 21.2  | 21.2  | 21.1  | 18.6  | 15.5  | 11.9  | 8.0  |
|                                                                                  |             | MCDB   | 9.7   | 11.1  | 14.7  | 21.1  | 24.1  | 28.0  | 28.3  | 28.4  | 24.2  | 18.5  | 13.6  | 9.6  |
|                                                                                  | 5%          | WB     | 6.2   | 7.2   | 9.5   | 13.4  | 17.0  | 19.8  | 20.4  | 20.1  | 17.6  | 14.4  | 10.5  | 6.7  |
|                                                                                  |             | MCDB   | 7.9   | 9.5   | 13.0  | 18.8  | 22.5  | 25.3  | 27.1  | 26.8  | 22.2  | 17.2  | 12.1  | 8.1  |
|                                                                                  | 10%         | WB     | 4.6   | 5.5   | 8.1   | 12.1  | 16.0  | 18.6  | 19.6  | 19.3  | 16.6  | 13.3  | 9.1   | 5.6  |
|                                                                                  |             | MCDB   | 6.0   | 7.6   | 11.0  | 17.2  | 21.0  | 23.8  | 25.6  | 25.2  | 20.6  | 15.7  | 10.4  | 6.8  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 5.1   | 6.4   | 8.3   | 10.6  | 10.6  | 10.4  | 10.7  | 11.1  | 9.7   | 7.8   | 5.4   | 4.7  |
|                                                                                  |             | MCDBR  | 6.6   | 8.7   | 12.8  | 14.6  | 14.2  | 14.1  | 14.7  | 14.6  | 13.3  | 10.6  | 7.7   | 6.0  |
|                                                                                  |             | MCWBR  | 5.1   | 6.1   | 7.7   | 7.7   | 7.0   | 6.5   | 5.7   | 5.9   | 6.5   | 6.4   | 5.6   | 4.7  |
|                                                                                  | 5% WB       | MCDBR  | 6.5   | 8.2   | 11.3  | 13.5  | 12.7  | 13.0  | 12.9  | 12.7  | 11.8  | 9.7   | 6.8   | 5.8  |
|                                                                                  |             | MCWBR  | 5.2   | 6.0   | 7.3   | 7.4   | 6.8   | 6.4   | 5.5   | 5.6   | 6.2   | 6.2   | 5.3   | 4.7  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.310  | 0.333 | 0.376 | 0.420 | 0.421 | 0.422 | 0.437 | 0.430 | 0.395 | 0.368 | 0.340 | 0.307 |      |
|                                                                                  | taud        | 2.352  | 2.321 | 2.252 | 2.174 | 2.212 | 2.245 | 2.209 | 2.245 | 2.310 | 2.359 | 2.386 | 2.388 |      |
|                                                                                  | Ebn at Noon | 725    | 799   | 821   | 821   | 838   | 839   | 818   | 803   | 793   | 747   | 681   | 676   |      |
|                                                                                  | Edh at Noon | 67     | 88    | 112   | 135   | 135   | 132   | 135   | 124   | 105   | 84    | 65    | 57    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 0.87   | 1.69  | 2.70  | 4.17  | 4.99  | 5.38  | 5.10  | 4.53  | 3.22  | 1.87  | 0.90  | 0.65  |      |
|                                                                                  | RadStd      | 0.15   | 0.24  | 0.42  | 0.55  | 0.57  | 0.44  | 0.56  | 0.45  | 0.43  | 0.27  | 0.11  | 0.07  |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) | Station Only            | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (47) | Regional (60 neighbors) | +0.57 | N/A     | N/A    | +0.81   | N/A   | N/A   | N/A         | N/A     | N/A     | +40     |
|      |                         | +0.48 | N/A     | N/A    | N/A     | +0.35 | +0.33 | N/A         | N/A     | +61     | +22     |

Nomenclature: See separate page